

ROCHELLE, IL, USA

WMO: 722182

Lat: 41.893N

Lon: 89.078W

Elev: 781

StdP: 14.29

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04890

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.4	(c) -2.3	(d) -15.0	(e) 2.5	(f) -8.0	(g) -8.9	(h) 3.5	(i) -0.8	(j) 28.7	(k) 25.3	(l) 26.5	(m) 24.6	(n) 6.0	(o) 270	(p) 0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	90.0	73.4	87.2	73.0	84.2	71.7	77.9	85.2	75.9	83.2	74.2	81.1	12.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	137.4	82.5	73.4	127.7	80.3	72.1	122.2	79.0	41.9	85.1	39.9	83.0	38.2	81.0	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.6	23.6	20.2	-18.4	93.4	9.5	2.6	-25.3	95.2	-30.9	96.7	-36.2	98.2	-43.2	100.0		
			-18.7	81.1	9.0	1.9	-25.2	82.5	-30.4	83.6	-35.4	84.7	-41.9	86.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	20.9	23.1	36.3	48.7	60.5	69.3	71.2	69.0	63.5	50.9	37.5	25.6
	DBStd	20.42	12.68	12.32	10.95	9.02	9.26	6.06	5.65	5.38	7.66	9.56	9.79	11.37
	HDD50	3505	901	754	446	131	15	0	0	0	2	107	390	759
	HDD65	6799	1366	1173	891	491	201	26	11	23	117	451	826	1223
	CDD50	2841	0	1	22	93	341	578	659	590	409	134	13	1
	CDD65	659	0	0	2	3	62	154	205	148	73	12	0	0
	CDH74	6214	0	0	18	73	708	1426	1888	1249	713	139	0	0
	CDH80	1794	0	0	2	6	242	449	568	294	202	31	0	0
	WSAvg	8.8	10.4	10.3	10.5	11.7	10.4	8.1	5.6	4.9	6.1	8.1	9.8	9.9
	PrecAvg	35.2	1.9	1.5	2.3	3.2	4.6	4.4	3.6	3.8	3.0	2.3	2.4	1.7
Precipitation	PrecMax	42.4	3.5	4.2	4.7	6.1	12.1	9.8	7.6	8.5	6.1	6.7	5.1	3.2
	PrecMin	22.6	0.5	0.5	0.9	1.3	0.7	0.8	1.1	0.0	0.6	0.3	0.2	0.4
	PrecStd	5.2	0.8	1.0	1.2	1.4	2.8	2.5	1.8	1.7	1.6	1.5	1.5	0.8
	PrecStd	5.2	0.8	1.0	1.2	1.4	2.8	2.5	1.8	1.7	1.6	1.5	1.5	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	59.1	76.6	80.6	91.8	91.7	92.9	90.3	90.2	84.0	68.5	58.4
		MCWB	49.6	53.1	64.1	60.6	70.1	72.9	79.2	74.6	73.1	69.7	56.8	56.2
	2%	DB	45.7	49.0	64.4	75.6	86.4	89.1	89.0	86.2	86.2	77.7	61.4	50.3
		MCWB	43.6	45.4	56.5	58.9	67.5	72.1	77.4	74.4	71.3	63.7	53.6	47.4
	5%	DB	39.4	41.6	58.4	70.9	81.8	85.6	86.0	82.9	81.5	72.3	56.7	44.5
		MCWB	37.2	38.8	51.9	57.1	64.8	71.3	74.8	72.9	67.9	60.6	51.5	42.6
	10%	DB	36.2	37.2	52.1	65.9	77.3	81.6	82.6	80.9	78.1	67.5	52.3	39.6
		MCWB	34.9	35.0	46.6	54.9	63.9	68.7	72.6	70.9	66.3	60.2	47.1	37.9
	0.4%	WB	50.9	53.0	64.4	65.6	73.7	79.1	82.2	78.9	76.6	71.9	60.3	56.2
		MCDB	51.6	57.7	75.3	72.6	83.3	87.1	90.3	86.2	84.5	81.8	64.5	58.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.9	45.3	58.5	62.0	71.1	75.4	78.9	76.5	74.0	67.7	55.7	47.2
		MCDB	45.5	48.5	63.9	70.3	81.3	83.4	86.3	83.5	81.6	72.7	60.5	48.5
	5%	WB	37.6	39.3	52.9	59.2	68.4	73.4	76.8	74.6	71.4	64.0	51.7	42.1
		MCDB	39.1	41.3	57.0	67.9	77.5	82.2	84.0	81.0	78.7	68.5	56.1	44.0
	10%	WB	34.8	35.4	46.8	55.8	66.0	71.4	74.8	72.7	69.2	60.1	47.6	37.9
		MCDB	36.1	37.3	51.9	65.1	74.3	78.7	81.1	78.4	75.1	66.3	52.0	39.5
Mean Daily Temperature Range	5% DB	MDBR	15.1	15.9	17.0	21.9	22.0	20.3	20.4	20.8	23.3	21.7	17.1	14.0
		MCDBR	17.0	19.2	25.3	30.2	27.4	24.2	22.4	22.6	26.1	28.0	24.0	18.8
		MCWBR	14.7	15.2	16.9	17.1	12.5	11.8	12.5	13.0	13.0	15.3	16.9	16.3
	5% WB	MCDBR	16.2	18.3	23.0	25.1	22.3	19.9	20.1	19.9	21.3	21.4	21.7	17.8
		MCWBR	14.4	15.0	16.6	16.4	12.2	11.5	12.5	12.5	11.9	15.1	16.7	16.1
Clear-Sky Solar Irradiance	taub		0.286	0.304	0.335	0.386	0.417	0.425	0.440	0.421	0.394	0.342	0.302	0.283
	taud		2.410	2.373	2.398	2.265	2.232	2.256	2.201	2.251	2.301	2.450	2.506	2.445
	Ebn at Noon		275	286	290	280	273	270	264	266	265	269	268	266
	Edh at Noon		25	30	33	41	44	43	45	41	36	28	23	22
	RadAvg		538	839	1162	1476	1709	1937	1930	1701	1421	934	606	445
All-Sky Solar Radiation	RadStd		62	104	102	123	122	145	111	102	121	110	52	48

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page